

1 Systems of Linear Equations



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1.1 Introduction to Systems of Linear Equations

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Objectives

- Recognize a linear equation in n variables.
- Find a parametric representation of a solution set.
- Determine whether a system of linear equations is consistent or inconsistent.
- Use back-substitution and Gaussian elimination to solve a system of linear equations.

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Linear Equations in n Variables

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Linear Equations in n Variables (1 of 1)

Definition of a Linear Equation in n Variables

A linear equation in n variables $x_1, x_2, x_3, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b.$$

The **coefficients** $a_1, a_2, a_3, \dots, a_n$ are real numbers, and the **constant term** b is a real number. The number a_1 is the **leading coefficient**, and x_1 is the **leading variable**.

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Example 1 – Linear and Nonlinear Equations

Each equation is linear.

a. $3x + 2y = 7$

b. $\frac{1}{2}x + y - \pi z = \sqrt{2}$

c. $(\sin \pi)x_1 - 4x_2 = e^2$

Each equation is not linear.

a. $xy + z = 2$

b. $e^x - 2y = 4$

c. $\sin x_1 + 2x_2 - 3x_3 = 0$

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Solutions and Solution Sets

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Solutions and Solution Sets (1 of 2)

A **solution** of a linear equation in n variables is a sequence of n real numbers $s_1, s_2, s_3, \dots, s_n$ that satisfy the equation when you substitute the values

$$x_1 = s_1, x_2 = s_2, x_3 = s_3, \dots, x_n = s_n$$

into the equation.

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Solutions and Solution Sets (2 of 2)

The set of *all* solutions of a linear equation is its **solution set**.

To describe the entire solution set of a linear equation, use a **parametric representation**.

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Example 3 – Parametric Representation of a Solution Set

Solve the linear equation $3x + 2y - z = 3$.

Solution:

Choosing y and z to be the free variables, solve for x to obtain

$$3x = 3 - 2y + z$$

$$x = 1 - \frac{2}{3}y + \frac{1}{3}z.$$

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Example 3 – Solution

Letting $y = s$ and $z = t$, you obtain the parametric representation

$$x = 1 - \frac{2}{3}s + \frac{1}{3}t, \quad y = s, \quad z = t$$

where s and t are any real numbers. Two particular solutions are

$$x = 1, y = 0, z = 0 \quad \text{and} \quad x = 1, y = 1, z = 2.$$

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Systems of Linear Equations

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Systems of Linear Equations (1 of 4)

A **system of m linear equations in n variables** is a set of m equations, each of which is linear in the same n variables:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m.$$

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Systems of Linear Equations (2 of 4)

A system of linear equations is also called a **linear system**. A **solution** of a linear system is a sequence of numbers $s_1, s_2, s_3, \dots, s_n$ that is a solution of each equation in the system.

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Example 4 – Systems of Two Equations in Two Variables

Solve and graph each system of linear equations.

a. $x + y = 3$
 $x - y = -1$

b. $x + y = 3$
 $2x + 2y = 6$

c. $x + y = 3$
 $x + y = 1$

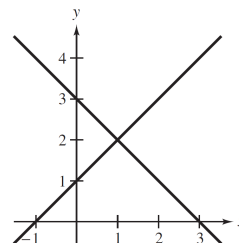
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Example 4 – Solution (1 of 4)

- a.** This system has exactly one solution, $x = 1$ and $y = 2$.
 One way to obtain the solution is to add the two equations to give $2x = 2$, which implies $x = 1$ and so $y = 2$.

The graph of this system is two *intersecting* lines, as shown in Figure 1.1(a).



- a.** Two intersecting lines:
 $x + y = 3$
 $x - y = -1$

Figure 1.1

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Example 4 – Solution (2 of 4)

- b.** This system has infinitely many solutions because the second equation is the result of multiplying both sides of the first equation by 2.

A parametric representation of the solution set is

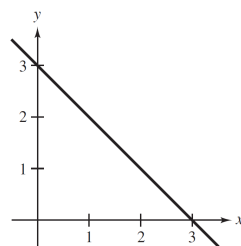
$$x = 3 - t, y = t, t \text{ is any real number.}$$

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Example 4 – Solution (3 of 4)

The graph of this system is two *coincident* lines, as shown in Figure 1.1(b).



- b.** Two coincident lines:

$$x + y = 3$$

$$2x + 2y = 6$$

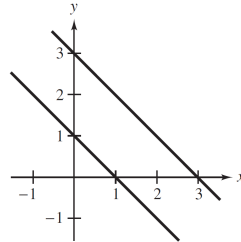
Figure 1.1

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Example 4 – Solution (4 of 4)

- c. This system has no solution because the sum of two numbers cannot be 3 and 1 simultaneously. The graph of this system is two *parallel* lines, as shown in Figure 1.1(c).



c. Two parallel lines:

$$x + y = 3$$

$$x + y = 1$$

Figure 1.1

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Systems of Linear Equations (3 of 4)

Number of Solutions of a System of Linear Equations

For a system of linear equations, precisely one of the statements below is true.

1. The system has exactly one solution (consistent system).
2. The system has infinitely many solutions (consistent system).
3. The system has no solution (inconsistent system).

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Example 7 – Using Elimination to Rewrite a System in Row-Echelon Form

Solve the system.

$$\begin{aligned}x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17\end{aligned}$$

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Example 7 – Solution (1 of 2)

Work from the upper left corner of the system, saving the x at the upper left and eliminating the other x -terms from the first column.

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2x - 5y + 5z &= 17\end{aligned}$$

← Adding the first equation to the second equation produces a new second equation.

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ -y - z &= -1\end{aligned}$$

← Adding -2 times the first equation to the third equation produces a new third equation.

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Example 7 – Solution (2 of 2)

Now that you have eliminated all but the first x from the first column, work on the second column.

$$x - 2y + 3z = 9$$

$$y + 3z = 5$$

$$2z = 4$$



Adding the second equation to the third equation produces a new third equation.

$$x - 2y + 3z = 9$$

$$y + 3z = 5$$

$$z = 2$$



Multiplying the third equation by $\frac{1}{2}$ produces a new third equation.

The solution is $x = 1$, $y = -1$, $z = 2$.

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Systems of Linear Equations (4 of 4)

Each of the three equations in Example 7 represents a plane in a three-dimensional coordinate system.

The unique solution of the system is the point $(x, y, z) = (1, -1, 2)$, so the three planes intersect at this point, as shown in Figure 1.2.

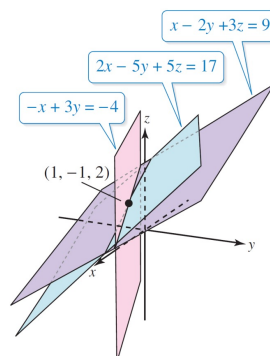


Figure 1.2

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1 Systems of Linear Equations



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1.2 Gaussian Elimination and Gauss-Jordan Elimination

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Objectives

- Determine the size of a matrix and write an augmented or coefficient matrix from a system of linear equations.
- Use matrices and Gaussian elimination with back-substitution to solve a system of linear equations.
- Use matrices and Gauss-Jordan elimination to solve a system of linear equations.
- Solve a homogeneous system of linear equations.

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Matrices

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Matrices (1 of 3)

Definition of a Matrix

If m and n are positive integers, then an $m \times n$ (read “ m by n ”) matrix is a rectangular array

	Column 1	Column 2	Column 3	...	Column n
Row 1	a_{11}	a_{12}	a_{13}	$\dots$	a_{1n}
Row 2	a_{21}	a_{22}	a_{23}	$\dots$	a_{2n}
Row 3	a_{31}	a_{32}	a_{33}	$\dots$	a_{3n}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
Row m	a_{m1}	a_{m2}	a_{m3}	$\dots$	a_{mn}

in which each **entry**, a_{ij} , of the matrix is a number.

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Matrices (2 of 3)

The entry a_{ij} is located in the i th row and the j th column. The index i is called the **row subscript** because it identifies the row in which the entry lies, and the index j is called the **column subscript** because it identifies the column in which the entry lies.

A matrix with m rows and n columns is of **size** $m \times n$. When $m = n$, the matrix is **square** of **order** n and the entries a_{11} , a_{22} , a_{33} , $\dots$, a_{nn} are the **main diagonal** entries.

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Example 1 – Sizes of Matrices

Each matrix has the indicated size.

a. $[2]$ Size: 1×1

b. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Size: 2×2

c. $\begin{bmatrix} e & 2 & -7 \\ \pi & \sqrt{2} & 4 \end{bmatrix}$ Size: 2×3

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Matrices (3 of 3)

The matrix derived from the coefficients and constant terms of a system of linear equations is the **augmented matrix** of the system. The matrix containing only the coefficients of the system is the **coefficient matrix** of the system.

System	Augmented Matrix	Coefficient Matrix
$\begin{array}{rcl} x - 4y + 3z & = & 5 \\ -x + 3y - z & = & -3 \\ 2x & - & 4z = 6 \end{array}$	$\left[\begin{array}{cccc} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{array} \right]$

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Elementary Row Operations

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Elementary Row Operations (1 of 4)

Elementary Row Operations

1. Interchange two rows.
2. Multiply a row by a nonzero constant.
3. Add a multiple of a row to another row.

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Example 3 – Using Elementary Row Operations to Solve a System (1 of 3)

Linear System

$$\begin{aligned}x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17\end{aligned}$$

Add the first equation to the second equation.

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2x - 5y + 5z &= 17\end{aligned}$$

Add -2 times the first equation to the third equation.

Associated Augmented Matrix

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{bmatrix}$$

Add the first row to the second row to produce a new second row.

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 2 & -5 & 5 & 17 \end{bmatrix} \quad R_2 + R_1 \rightarrow R_2$$

Add -2 times the first row to the third row to produce a new third row.

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Example 3 – Using Elementary Row Operations to Solve a System (2 of 3)

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ -y - z &= -1\end{aligned}$$

Add the second equation to the third equation.

$$\begin{aligned}x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2z &= 4\end{aligned}$$

Multiply the third equation by $\frac{1}{2}$.

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{bmatrix} \quad R_3 + (-2)R_1 \rightarrow R_3$$

Add the second row to the third row to produce a new third row.

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix} \quad R_3 + R_2 \rightarrow R_3$$

Multiply the third row by $\frac{1}{2}$ to produce a new third row.

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Example 3 – Using Elementary Row Operations to Solve a System (3 of 3)

$$\begin{array}{rcl} x - 2y + 3z = 9 \\ y + 3z = 5 \\ z = 2 \end{array} \quad \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad \left(\frac{1}{2}\right)R_3 \rightarrow R_3$$

Use back-substitution to find that the solution is $x = 1$, $y = -1$, and $z = 2$.

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Elementary Row Operations (2 of 4)

Row-Echelon Form and Reduced Row-Echelon Form

A matrix in **row-echelon form** has the properties below.

1. Any rows consisting entirely of zeros occur at the bottom of the matrix.
2. For each row that does not consist entirely of zeros, the first nonzero entry is 1 (called a **leading 1**).
3. For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row.

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Elementary Row Operations (3 of 4)

Row-Echelon Form and Reduced Row-Echelon Form

A matrix in row-echelon form is in **reduced row-echelon form** when every column that has a leading 1 has zeros in every position above and below its leading 1.

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Elementary Row Operations (4 of 4)

The procedure for using Gaussian elimination with back-substitution is summarized below.

Gaussian Elimination with Back-Substitution

1. Write the augmented matrix of the system of linear equations.
2. Use elementary row operations to rewrite the matrix in row-echelon form.
3. Write the system of linear equations corresponding to the matrix in row-echelon form, and use back-substitution to find the solution.

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Example 5 – Gaussian Elimination with Back-Substitution

Solve the system.

$$x_2 + x_3 - 2x_4 = -3$$

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 + 4x_2 + x_3 - 3x_4 = -2$$

$$x_1 - 4x_2 - 7x_3 - x_4 = -19$$

Solution:

The augmented matrix for this system is

$$\left[\begin{array}{ccccc} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

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Example 5 – Solution (1 of 3)

Obtain a leading 1 in the upper left corner and zeros elsewhere in the first column.

$$\left[\begin{array}{ccccc} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

Interchange the first two rows.

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccccc} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

Adding -2 times the first row to the third row produces a new third row.

$$R_3 + (-2)R_1 \rightarrow R_3$$

$$\left[\begin{array}{ccccc} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & -6 & -6 & -1 & -21 \end{array} \right]$$

Adding -1 times the first row to the fourth row produces a new fourth row.

$$R_4 + (-1)R_1 \rightarrow R_4$$

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Example 5 – Solution (2 of 3)

Now that the first column is in the desired form, change the second column as shown below.

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & 0 & 0 & -13 & -39 \end{bmatrix}$$

Adding 6 times the second row to the fourth row produces a new fourth row. $R_4 + (6)R_2 \rightarrow R_4$

To write the third and fourth columns in proper form, multiply the third row by $\frac{1}{3}$ and the fourth row by $-\frac{1}{13}$.

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Multiplying the third row by $\frac{1}{3}$ and the fourth row by $-\frac{1}{13}$ produces new third and fourth rows. $(\frac{1}{3})R_3 \rightarrow R_3$
 $(-\frac{1}{13})R_4 \rightarrow R_4$

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Example 5 – Solution (3 of 3)

The matrix is now in row-echelon form, and the corresponding system is shown below.

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 2 \\ x_2 + x_3 - 2x_4 &= -3 \\ x_3 - x_4 &= -2 \\ x_4 &= 3 \end{aligned}$$

Use back-substitution to find that the solution is $x_1 = -1$, $x_2 = 2$, $x_3 = 1$, and $x_4 = 3$.

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Gauss-Jordan Elimination

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Gauss-Jordan Elimination (1 of 1)

With Gaussian elimination, you apply elementary row operations to a matrix to obtain a (row-equivalent) row-echelon form. A second method of elimination, called **Gauss-Jordan elimination** after Carl Friedrich Gauss and Wilhelm Jordan (1842–1899), continues the reduction process until a *reduced* row-echelon form is obtained.

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Example 7 – Gauss-Jordan Elimination

Use Gauss-Jordan elimination to solve the system.

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

Solution:

You used Gaussian elimination to obtain the row-echelon form

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

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Example 7 – Solution (1 of 2)

Now, apply elementary row operations until you obtain zeros above each of the leading 1's, as shown below.

$$\begin{bmatrix} 1 & 0 & 9 & 19 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$R_1 + (2)R_2 \rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 9 & 19 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$R_2 + (-3)R_3 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$R_1 + (-9)R_3 \rightarrow R_1$$

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Example 7 – Solution (2 of 2)

The matrix is now in reduced row-echelon form. Converting back to a system of linear equations, you have

$$x = 1$$

$$y = -1$$

$$z = 2.$$

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Homogeneous Systems of Linear Equations

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Homogeneous Systems of Linear Equations (1 of 2)

Systems of linear equations in which each of the constant terms is zero are called **homogeneous**. A homogeneous system of m equations in n variables has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= 0. \end{aligned}$$

A homogeneous system must have at least one solution. Specifically, if all variables in a homogeneous system have the value zero, then each of the equations is satisfied. Such a solution is **trivial** (or obvious).

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Example 9 – Solving a Homogeneous System of Linear Equations

Solve the system of linear equations.

$$x_1 - x_2 + 3x_3 = 0$$

$$2x_1 + x_2 + 3x_3 = 0$$

Solution:

Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix}$$

yields the matrices shown below.

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Example 9 – Solution (1 of 2)

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{bmatrix} \quad R_2 + (-2)R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \quad \left(\frac{1}{3}\right)R_2 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \quad R_1 + R_2 \rightarrow R_1$$

The system of equations corresponding to this matrix is

$$\begin{aligned} x_1 + 2x_3 &= 0 \\ x_2 - x_3 &= 0. \end{aligned}$$

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Example 9 – Solution (2 of 2)

Using the parameter $t = x_3$, the solution set is $x_1 = -2t$, $x_2 = t$, and $x_3 = t$, where t is any real number. This system has infinitely many solutions, one of which is the trivial solution ($t = 0$).

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Homogeneous Systems of Linear Equations (2 of 2)

Theorem 1.1 The Number of Solutions of a Homogeneous System

Every homogeneous system of linear equations is consistent. Moreover, if the system has fewer equations than variables, then it must have infinitely many solutions.

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1 Systems of Linear Equations



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1.3

Applications of Systems of Linear Equations

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Objectives

- Set up and solve a system of equations to fit a polynomial function to a set of data points.
- Set up and solve a system of equations to represent a network.

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Applications of Systems of Linear Equations

Systems of linear equations arise in a wide variety of applications. In this section you will look at two applications.

The first application shows how to fit a polynomial function to a set of data points in the plane.

The second application focuses on networks and Kirchhoff's Laws for electricity.

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Polynomial Curve Fitting

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Polynomial Curve Fitting (1 of 2)

Consider n points in the xy -plane

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

that represent a collection of data, and you want to find a polynomial function of degree $n - 1$

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1}$$

whose graph passes through the points.

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Polynomial Curve Fitting (2 of 2)

This procedure is called **polynomial curve fitting**. When all x -coordinates of the points are distinct, there is precisely one polynomial function of degree $n - 1$ (or less) that fits the n points, as shown in Figure 1.3.

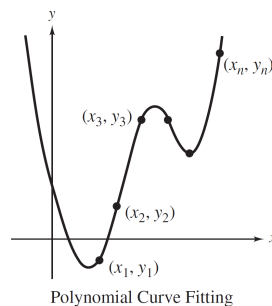


Figure 1.3

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Example 1 – Polynomial Curve Fitting

Determine the polynomial $p(x) = a_0 + a_1x + a_2x^2$ whose graph passes through the points (1, 4), (2, 0), and (3, 12).

Solution:

Substituting $x = 1, 2$, and 3 into $p(x)$ and equating the results to the respective y -values produces the system of linear equations in the variables a_0 , a_1 , and a_2 shown below.

$$p(1) = a_0 + a_1(1) + a_2(1)^2 = a_0 + a_1 + a_2 = 4$$

$$p(2) = a_0 + a_1(2) + a_2(2)^2 = a_0 + 2a_1 + 4a_2 = 0$$

$$p(3) = a_0 + a_1(3) + a_2(3)^2 = a_0 + 3a_1 + 9a_2 = 12$$

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Example 1 – Solution (1 of 2)

The solution of this system is

$$a_0 = 24, a_1 = -28, \text{ and } a_2 = 8$$

so the polynomial function is

$$p(x) = 24 - 28x + 8x^2.$$

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Example 1 – Solution (2 of 2)

Figure 1.4 shows the graph of p .

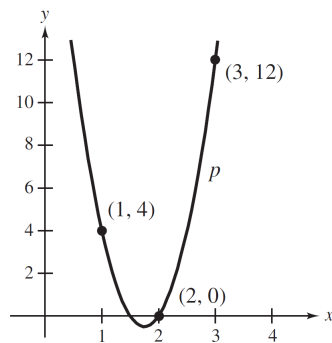


Figure 1.4

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Example 4 – An Application of Curve Fitting

Find a polynomial that relates the periods of the three planets that are closest to the Sun to their mean distances from the Sun, as shown in the table. Then use the polynomial to calculate the period of Mars, and compare it to the value shown in the table. (The mean distances are in astronomical units, and the periods are in years.)

Planet	Mercury	Venus	Earth	Mars
Mean Distance	0.387	0.723	1.000	1.524
Period	0.241	0.615	1.000	1.881

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Example 4 – Solution (1 of 3)

Begin by fitting a quadratic polynomial function

$$p(x) = a_0 + a_1x + a_2x^2$$

to the points

(0.387, 0.241), (0.723, 0.615), and (1, 1).

The system of linear equations obtained by substituting these points into $p(x)$ is

$$\begin{aligned} a_0 + 0.387a_1 + (0.387)^2a_2 &= 0.241 \\ a_0 + 0.723a_1 + (0.723)^2a_2 &= 0.615 \\ a_0 + a_1 + a_2 &= 1. \end{aligned}$$

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Example 4 – Solution (2 of 3)

The approximate solution of the system is

$$a_0 \approx -0.0634, a_1 \approx 0.6119, a_2 \approx 0.4515$$

which means that an approximation of the polynomial function is

$$p(x) = -0.0634 + 0.6119x + 0.4515x^2.$$

Using $p(x)$ to evaluate the period of Mars produces

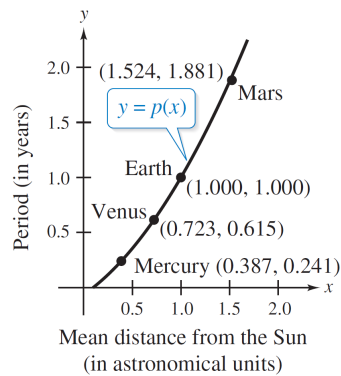
$$p(1.524) \approx 1.918 \text{ years.}$$

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Example 4 – Solution (3 of 3)

Note that the period of Mars is shown in the table as 1.881 years. The figure below provides a graphical comparison of the polynomial function to the values shown in the table.



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Network Analysis

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Network Analysis (1 of 3)

Networks composed of branches and junctions are used as models in such fields as economics, traffic analysis, and electrical engineering.

In a network model, you assume that the total flow into a junction is equal to the total flow out of the junction.

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Example 5 – Analysis of a Network

Set up a system of linear equations to represent the network shown in Figure 1.6. Then solve the system.

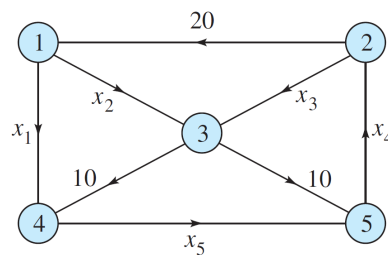


Figure 1.6

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Example 5 – Solution (1 of 3)

Each of the network's five junctions gives rise to a linear equation, as shown below.

$$x_1 + x_2 = 20 \quad \text{Junction 1}$$

$$x_3 - x_4 = -20 \quad \text{Junction 2}$$

$$x_2 + x_3 = 20 \quad \text{Junction 3}$$

$$x_1 - x_5 = -10 \quad \text{Junction 4}$$

$$-x_4 + x_5 = -10 \quad \text{Junction 5}$$

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Example 5 – Solution (2 of 3)

The augmented matrix for this system is

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 0 & 0 & -1 & 1 & -10 \end{bmatrix}.$$

Gauss-Jordan elimination produces the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 1 & 0 & 0 & 1 & 30 \\ 0 & 0 & 1 & 0 & -1 & -10 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

From the matrix above,

$$x_1 - x_5 = -10, \quad x_2 + x_5 = 30, \quad x_3 - x_5 = -10, \quad \text{and} \quad x_4 - x_5 = 10.$$

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Example 5 – Solution (3 of 3)

Letting $t = x_5$, you have

$$x_1 = t - 10, \quad x_2 = -t + 30, \quad x_3 = t - 10, \quad x_4 = t + 10, \quad x_5 = t$$

where t is any real number, so this system has infinitely many solutions.

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Network Analysis (2 of 3)

An electrical network is another type of network where analysis is commonly applied. An analysis of such a system uses two properties of electrical networks known as **Kirchhoff's Laws**.

1. All the current flowing into a junction must flow out of it.
2. The sum of the products IR (I is current and R is resistance) around a closed path is equal to the total voltage in the path.

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Network Analysis (3 of 3)

In an electrical network, current is measured in amperes, or amps (A), resistance is measured in ohms (Ω , the Greek letter omega), and the product of current and resistance is measured in volts (V).

The symbol $\text{---}||\text{---}$ represents a battery. The larger vertical bar denotes where the current flows out of the terminal. The symbol $\text{---}\wedge\wedge\wedge\text{---}$ denotes resistance. An arrow in the branch shows the direction of the current.

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Example 6 – Analysis of an Electrical Network

Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.7.

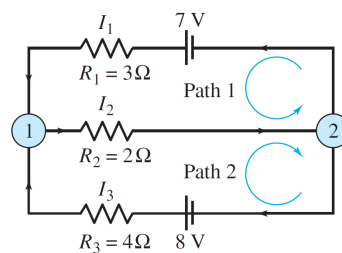


Figure 1.7

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Example 6 – Solution (1 of 3)

Applying Kirchhoff's first law to either junction produces

$$I_1 + I_3 = I_2 \quad \text{Junction 1 or Junction 2}$$

and applying Kirchhoff's second law to the two paths produces

$$R_1 I_1 + R_2 I_2 = 3I_1 + 2I_2 = 7 \quad \text{Path 1}$$

$$R_2 I_2 + R_3 I_3 = 2I_2 + 4I_3 = 8. \quad \text{Path 2}$$

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Example 6 – Solution (2 of 3)

So, you have the system of three linear equations in the variables I_1 , I_2 , and I_3 shown below.

$$I_1 - I_2 + I_3 = 0$$

$$3I_1 + 2I_2 = 7$$

$$2I_2 + 4I_3 = 8$$

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Example 6 – Solution (3 of 3)

Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 7 \\ 0 & 2 & 4 & 8 \end{bmatrix}$$

produces the reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

which means $I_1 = 1$ amp, $I_2 = 2$ amps, and $I_3 = 1$ amp.

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